

MHA (with KV-encoding)

• store K, V for all seq, and only add new token.

Dimensions:

$$\begin{aligned} X_{\text{new}} &\in \mathbb{R}^{(1, d_m)} \\ W_Q &\in \mathbb{R}^{(d_m, d_m)} \\ W_K &\in \mathbb{R}^{(d_m, d_m)} \\ W_V &\in \mathbb{R}^{(d_m, d_m)} \end{aligned}$$

seq_pos = current sequence generated.

$M = \text{nr heads}$

$d_m = \text{embedding dimension}$.

① Calculate Q (for 1 token), don't calculate K, V (only 1 time)

$$Q_{\text{new}} = X_{\text{new}} \cdot W_Q$$

$$\begin{aligned} X_{\text{new}} &\in \mathbb{R}^{(1, d_m)} \\ W_Q &\in \mathbb{R}^{(d_m, d_m)} \Rightarrow Q_{\text{new}} \in \mathbb{R}^{(1, d_m)} \end{aligned}$$

$$\begin{aligned} \frac{\text{Computational - Cost}}{Q_{\text{new}}} &= (1, d_m) \cdot (1, d_m) = \\ &= (1 \cdot d_m \cdot d_m) = \boxed{O(d_m^2)} \end{aligned}$$

$$\frac{\text{Memory - Cost}}{Q_{\text{new}}} = \boxed{O(d_m)}$$

• KV-cache (store buffer and add new token).

$$\begin{aligned} K'_{\text{cache}} &= \text{concatenate}(K_{\text{cache}}, K_{\text{new}}) \\ V'_{\text{cache}} &= \text{concatenate}(V_{\text{cache}}, V_{\text{new}}) \end{aligned}$$

• Dimensions:

$$\begin{aligned} K_{\text{cache}} &\in \mathbb{R}^{(\text{seq}_{\text{past}}, d_m)} \\ V_{\text{cache}} &\in \mathbb{R}^{(\text{seq}_{\text{past}}, d_m)} \end{aligned}$$

$$K_{\text{new}} \in \mathbb{R}^{(1, d_m)}$$

$$V_{\text{new}} \in \mathbb{R}^{(1, d_m)}$$

• after adding new token:

$$\begin{aligned} K'_{\text{cache}} &\in \mathbb{R}^{(\text{seq}_{\text{past}} + 1, d_m)} \\ V'_{\text{cache}} &\in \mathbb{R}^{(\text{seq}_{\text{past}} + 1, d_m)} \end{aligned}$$

• Computational cost: $\boxed{O(d_m)}$

→ linear because we keep concatenated.

• Memory cost K_{cache} and V_{cache} : $O(\text{seq}_{\text{past}}, d_m)$

③ Calculate Attn matrix $Q_{row} \cdot K_{cache}^T$

$$\text{Attention}_{\text{weights}} = \frac{Q_{row} \cdot K_{cache}^T}{\sqrt{d_k}}$$

• Dimensions:

- $Q_{row} \in \mathbb{R}^{(1, d_m)}$
- $K_{cache}^T \in \mathbb{R}^{(d_m, \text{res post})}$
- $Q_{row} \cdot K_{cache}^T \in \mathbb{R}^{(1, \text{res post})}$

Computational Cost: $\boxed{O(\text{res post} \cdot d_m)}$

$$(1, d_m) \cdot (d_m, \text{res post}) = (1 \cdot d_m \cdot \text{res post})$$

Memory Cost: $\boxed{O(\text{res post})}$

④ Attention rows $\cdot V_{cache}$

$$\text{Attn output} = \text{Attn weights} \cdot V_{cache}$$

$$\begin{aligned} \text{Attn weights} &\in \mathbb{R}^{(1, \text{res post})} \\ V_{cache} &\in \mathbb{R}^{(\text{res post}, d_m)} \\ \Rightarrow \text{Attn output} &\in \mathbb{R}^{(1, d_m)} \end{aligned}$$

• Computational cost: $\boxed{O(\text{res post} \cdot d_m)}$

• Memory cost: $\boxed{O(d_m)}$

• Final Projection W^o

$$\text{Output} = \text{Att}_{\text{output}} \cdot W^o$$

$$\text{dim} \cdot W^o \in \mathbb{R}^{(d_m, d_m)}$$

$$\text{Computational-cost} = \boxed{O(d_m^2)}$$

$$\text{memory-cost} = O(d_m)$$

	Computational-cost	memory-cost
Projection Q for new token	$O(d_m^2)$	$O(d_m)$
Add K, V in cache	$O(d_m)$	$O(\text{seq}_{\text{past}} \cdot d_m)$
$Q \cdot K^T$	$O(\text{seq}_{\text{past}} \cdot d_m)$	$O(\text{seq}_{\text{past}})$
Attention $\cdot V$ weight	$O(\text{seq}_{\text{past}} \cdot d_m)$	$O(d_m)$
Final Projection W^o	$O(d_m^2)$	$O(d_m)$

Final Conclusion

$$\text{Computational-cost} = \boxed{O(2d_m^2 + 2\text{seq}_{\text{past}} \cdot d_m + d_m)}$$

$$\text{Memory-cost} = \boxed{O(d_m^2 + \text{seq}_{\text{past}} \cdot d_m)}$$

(!) we move from $O(\text{seq}^2 d_m)$ to $O(\text{seq}_{\text{past}}, d_m)$ with KV-cache.